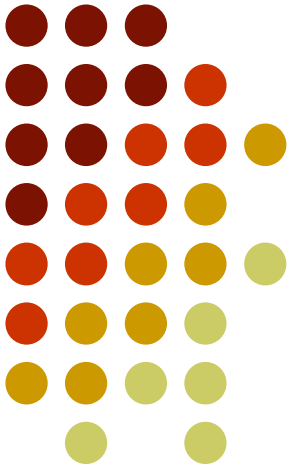
Hedging Strategies Using Futures

Chapter 3



Long & Short Hedges



- A long futures hedge is appropriate when you know you will purchase an asset in the future and want to lock in the price
- A short futures hedge is appropriate when you know you will sell an asset in the future and want to lock in the price

Example: A Short Hedge

- It is May 15. An oil producer has negotiated a contract to sell 1 million barrels of crude oil. The price in the sales contract is the spot price on August 15. Quotes:
  - Spot price of crude oil: $80 per barrel
  - August oil futures price: $79 per barrel

Example: A Long Hedge



- It is January 15. A copper fabricator knows it will require 100,000 pounds of copper on May 15 to meet a certain contract. The spot price of copper is 340 cents per pound and the May futures price is 320 cents per pound.

Arguments in Favor of Hedging



Companies should focus on the main business they are in and take steps to minimize risks arising from interest rates, exchange rates, and other market variables

Arguments against Hedging



- Shareholders are usually well diversified and can make their own hedging decisions
- It may increase risk to hedge when competitors do not
- Explaining a situation where there is a loss on the hedge and a gain on the underlying can be difficult

Example: Hedging and Competitors

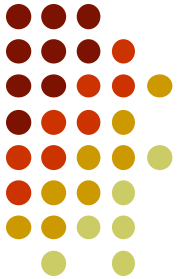
Table 3.1 Danger in hedging when competitors do not hedge.

Change in gold price	*Effect on price of gold jewelry*	*Effect on profits of TakeaChance Co.*	*Effect on profits of SafeandSure Co.*
Increase	Increase	None	Increase
Decrease	Decrease	None	Decrease

Discussion

- A corn farmer argues: “I do not use futures contracts for hedging. My real risk is not the price of corn. It is that my whole crop gets wiped out by the weather.” Discuss this viewpoint. Should the farmer estimate his or her expected production of corn and hedge to try to lock in a price for expected production?

Basis Risk



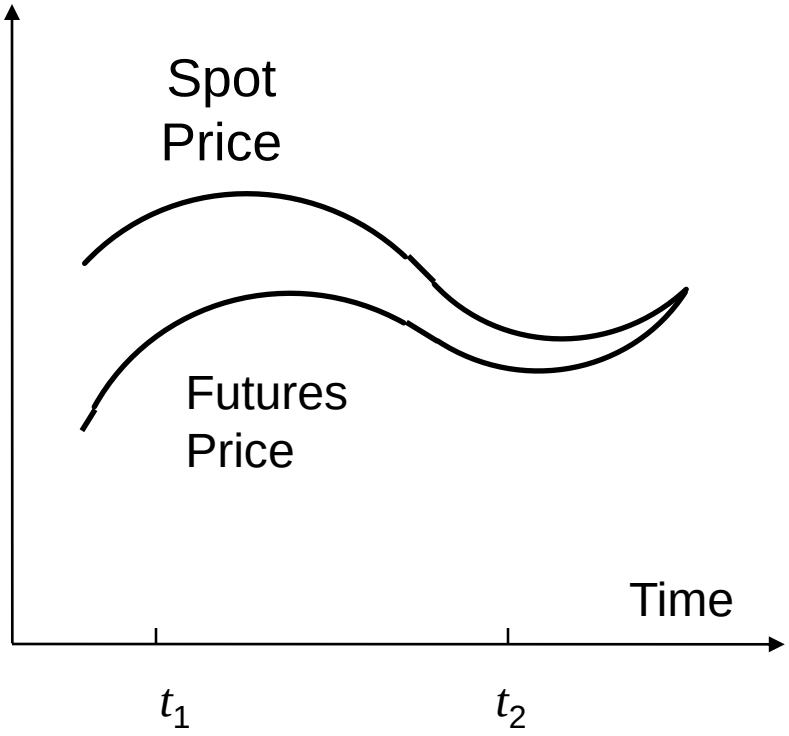
- Basis is the difference between spot & futures

Basis = Spot price of asset to be hedged – Futures price of contract used

- Basis risk arises because of the uncertainty about the basis when the hedge is closed out

Convergence of Futures to Spot

(Hedge initiated at time t_1 and closed out at time t_2; Figure 3.1, page 56)



Long Hedge for Purchase of an Asset

- Define

  F_1 : Futures price at time hedge is set up

  F_2 : Futures price at time asset is purchased

  S_2 : Asset price at time of purchase

  b_2 : Basis at time of purchase

Cost of asset	S_2
Gain on Futures	$F_2 - F_1$
Net amount paid	$S_2 - (F_2 - F_1) = F_1 + b_2$

Short Hedge for Sale of an Asset

- Define

  F_1 : Futures price at time hedge is set up

  F_2 : Futures price at time asset is sold

  S_2 : Asset price at time of sale

  b_2 : Basis at time of sale

Price of asset	S_2
Gain on Futures	$F_1 - F_2$
Net amount received	$S_2 + (F_1 - F_2) = F_1 + b_2$

Choice of Contract



- Choose a delivery month that is as close as possible to, but later than, the end of the life of the hedge
- When there is no futures contract on the asset being hedged, choose the contract whose futures price is most highly correlated with the asset price.

Example: Basis Risk in a Short Hedge



- It is March 1. A U.S. company expects to receive 50 million Japanese yen at the end of July. The September futures price for the yen is currently 1.2800 cents per yen. Its hedging strategy is:
  - Short four September yen futures contracts on March 1 at a futures price of 1.2800.
  - Close out the contract at the end of July.

  Suppose that the spot and futures prices when the contract is closed out are 1.2200 and 1.2250 cents per yen.

Example: Basis Risk in a Long Hedge



- It is June 8. A company knows that it will need to purchase 20,000 barrels of crude oil some time in October or November. The current December oil futures price is $88.00 per barrel. Its hedging strategy is:
  - Take a long position in 20 December oil futures contracts on June 8 at a futures price of $88
  - Close out the contract when ready to purchase the oil

  Suppose that the spot prices and futures price on November 10 are $95 per barrel and $92 per barrel, respectively.

Optimal Hedge Ratio

Proportion of the exposure that should optimally be hedged is

$$h = \rho \frac{\sigma_S}{\sigma_F}$$

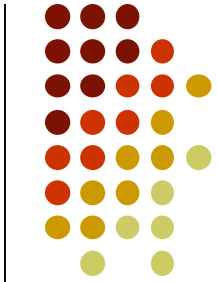
where

σ_S is the standard deviation of ΔS, the change in the spot price during the hedging period,

σ_F is the standard deviation of ΔF, the change in the futures price during the hedging period

ρ is the coefficient of correlation between ΔS and ΔF.

Example: Minimum Variance Hedge Ratio



- An airline expects to purchase two million gallons of jet fuel in one month and decides to use heating oil futures for hedging. Since jet fuel futures are not actively traded, it might choose to use heating oil futures contracts to hedge its exposure. Suppose we have data in the following table for 15 successive months.

Table 3.2 Data to calculate minimum variance hedge ratio when heating oil futures contract is used to hedge purchase of jet fuel.

Month i	*Change in heating oil futures price per gallon* $(=\Delta F)$	*Change in jet fuel price per gallon* $(=\Delta S)$
1	0.021	0.029
2	0.035	0.020
3	−0.046	−0.044
4	0.001	0.008
5	0.044	0.026
6	−0.029	−0.019
7	−0.026	−0.010
8	−0.029	−0.007
9	0.048	0.043
10	−0.006	0.011
11	−0.036	−0.036
12	−0.011	−0.018
13	0.019	0.009
14	−0.027	−0.032
15	0.029	0.023

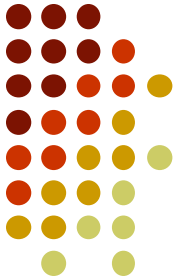
Example (Page 62)



- Airline will purchase 2 million gallons of jet fuel in one month and hedges using heating oil futures
- From historical data σ_F =0.0313, σ_S = 0.0263, and ρ= 0.928

$$h^* = 0.928 \times \frac{0.0263}{0.0313} = 0.7777$$

Optimal Number of Contracts



Q_A	Size of position being hedged (units)
Q_F	Size of one futures contract (units)
V_A	Value of position being hedged (=spot price time Q_A)
V_F	Value of one futures contract (=futures price times Q_F)

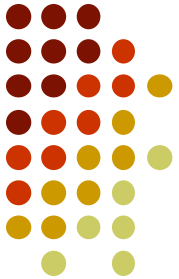
Optimal number of contracts if no tailing adjustment

$$= \frac{h^* Q_A}{Q_F}$$

Optimal number of contracts after tailing adjustment to allow for daily settlement of futures

$$= \frac{h^* V_A}{V_F}$$

Example continued



- The size of one heating oil contract is 42,000 gallons
- The spot price is 1.94 and the futures price is 1.99 (both dollars per gallon) so that

$$V_A = 1.94 \times 2{,}000{,}000 = 3{,}880{,}000$$
$$V_F = 1.99 \times 42{,}000 = 83{,}580$$

- Optimal number of contracts assuming no daily settlement $= 0.7777 \times 2{,}000{,}000/42{,}000 = 37.03$
- Optimal number of contracts after tailing

$$= 0.7777 \times 3{,}880{,}000/83{,}580 = 36.10$$

Hedging Using Index Futures
(Page 65)

To hedge the risk in a portfolio the number of contracts that should be shorted is

$$\beta \frac{V_A}{V_F}$$

where V_A is the current value of the portfolio, β is its beta, and V_F is the current value of one futures (=futures price times contract size)

Example

Futures price of S&P 500 is 1,010

Size of portfolio is $5.05 million

Beta of portfolio is 1.5

One contract is on $250 times the index

What position in futures contracts on the S&P 500 is necessary to hedge the portfolio?

Changing Beta



- What position is necessary to reduce the beta of the portfolio to 0.75?
- What position is necessary to increase the beta of the portfolio to 2.0?

Why Hedge Equity Returns

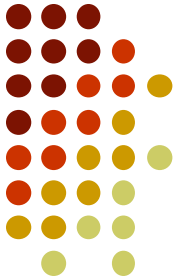


- May want to be out of the market for a while. Hedging avoids the costs of selling and repurchasing the portfolio
- If you feel the stocks in your portfolio have been chosen well and will outperform the market in both good and bad times. Hedging ensures that you will make money.

Stock Picking

- If you think you can pick stocks that will outperform the market, futures contract can be used to hedge the market risk
- If you are right, you will make money whether the market goes up or down

Stack and Roll (page 69-70)



- We can roll futures contracts forward to hedge future exposures
- Initially we enter into futures contracts to hedge exposures up to a time horizon
- Just before maturity we close them out and replace them with new contract reflect the new exposure

Example: Stack and Roll

- Suppose that in April 2014 a company realizes that it will have 1 million barrels of oil to sell in June 2015 and decides to hedge its risk with a hedge ratio of 1.0.

Table 3.5 Data for the example on rolling oil hedge forward.

Date	*Apr. 2014*	*Sept. 2014*	*Feb. 2015*	*June 2015*
Oct. 2014 futures price	88.20	87.40		
Mar. 2015 futures price		87.00	86.50	
July 2015 futures price			86.30	85.90
Spot price	89.00			86.00

Summary

- Long & Short hedges
  - A long futures hedge is appropriate when you know you will _____ an asset in the future.
  - A short futures hedge is appropriate when you know you will _____ an asset in the future.
- Basis risk
  - Basis = Spot price of asset to be hedged – Futures price of contract used
  - Basis risk arises because of the uncertainty about the basis when the hedge is closed out.
    - Long hedge: Net amount paid = F_1 + basis
    - Short hedge: Net amount received = F_1 + basis

Summary

- Choice of futures contract
  - Choose a delivery month that is as close as possible, but later than, the end of the life of the hedge
  - When there is no futures contract on the asset being hedged, choose the contract whose futures price is most highly correlated with the asset price
    - Optimal hedge ratio

$$h = \rho \frac{\sigma_S}{\sigma_F}$$

    - Optimal number of contracts

$$= \frac{h^* Q_A}{Q_F} \text{ (No tailing adjustment) } = \frac{h^* V_A}{V_F} \text{ (after tailing adjustment)}$$

Summary

- Hedging using index futures
  - Optimal number of contracts = $\beta \frac{V_A}{V_F}$
  - Changing beta
    - A short (long) position is required if beta increases (decreases)
- Stack and roll
  - When the expiration date of the hedge is later than the delivery dates of all the futures contracts that can be used, we can roll futures contracts forward.